China Western Math Olympiad 2024 — P1/4

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For positive integer n, denote $S_n = 1^{2024} + 2^{2024} + \cdots + n^{2024}$.

Prove that there exists infinitely many positive integers n, such that S_n isn't divisible by 1865 but S_{n+1} is divisible by 1865.

"All constructive problems are trivial."

Solution

We will show that if n is of the form $n = 1865^2\lambda - 2$ for $\lambda \in \mathbb{N}$ then $S_n \not\equiv 0 \pmod{1865}$ whilst $S_{n+1} \equiv 0 \pmod{1865}$.

Let $n = 1865^2\lambda - 2$. Notice that $S_{n+1} = S_n + (n+1)^{2024} \equiv S_n + (1865^2\lambda - 1)^{2024} \equiv S_n + 1 \pmod{1865}$ and so S_n and S_{n+1} aren't congruent modulo 1865. Denote $f(n) = 1865n$ over the non-negative integers. Notice that $f(n) \equiv 0 \pmod{1865}$. Denote $\alpha = 2024$. Consider,

$$\begin{aligned}
S_{n+1} &= ([f(0)+1]^\alpha + [f(0)+2]^\alpha + \cdots + [f(0)+1865]^\alpha) + \\
&+ ([f(1)+1]^\alpha + [f(1)+2]^\alpha + \cdots + [f(1)+1865]^\alpha) + \\
&+ \cdots \\
&+ ([f(1865\lambda-2)+1]^\alpha + [f(1865\lambda-2)+2]^\alpha + \cdots + [f(1865\lambda-2)+1865]^\alpha) \\
&+ ([f(1865\lambda-1)+1]^\alpha + [f(1865\lambda-1)+2]^\alpha + \cdots + [f(1865\lambda-1)+1864]^\alpha) \\
&\equiv (1865\lambda-1)S_{1865} + (S_{1865} - 0) \\
&\equiv (1865\lambda)S_{1865} \equiv 0 \pmod{1865}
\end{aligned}$$

Now since we already showed that $S_n \not\equiv S_{n+1} \pmod{1865}$ we are done. ■

Remark: The result can clearly be generalised, we leave it as an excercise to the reader.